

Representation	Selection	Reasoning
Is the evidence in the input?	Is the right evidence in context?	Is the evidence used correctly?
Modality ceiling two inline icons, parsed as FINDING <i>Colour exists only in pixels; a text input leaves the reasoner nothing to read.</i> Q What are the colors of the icons that the users touch to move the app up and remove the app respectively? Gold gray, red T ✗ “The document does not specify the colors” TL ✗ “it does not mention any visual details” ▼ add the page image V ✓ “The icon is gray. ... The icon is red.” TLV ✓ “gray (a gray upward arrow) ... red” DESIGN Start text-only; add the page image only when the question asks about appearance.	Evidence coverage author list on pages 13, 14 FINDING <i>Each page is relevant, yet neither suffices: the answer is a property of the page set.</i> Q How many chapter authors of this report come from Columbia University? Gold 3 page 13 ✗ “two” (Horton, Kinney) page 14 ✗ “one” (Garcia-Pando) ▼ add the complementary page both pages ✓ “three” (Horton, Kinney, Garcia-Pando) DESIGN Select evidence as a set: judge sufficiency of the whole set, not relevance page by page.	Evidence integration a diagram, an ablation table FINDING <i>Each size breaks a different step of one chain: locate the table, read the cell, compute.</i> Q If I drop the green module shown in Figure 6, what is the absolute percentage drop on 100-shot TACREV setting? Gold 0.5% 2B ✗ “2.4%” (reads another table) 4B ✗ “−9.6%” (right table, wrong row) 8B ✗ “(75.9 − 74.9) / 75.9 ... ≈ 1.32%” ▼ a larger reasoner 32B ✓ “75.9 − 75.4 = 0.5” DESIGN Match reasoner capacity to the operation; escalate questions that join a figure and a table.
Conversion fidelity scanned slide, text by parser FINDING <i>The error sits in the transcript, not the reasoner, and each parser fails differently.</i> Q What is the name of the company that created this presentation? Gold AVALAUNCH MEDIA PaddleOCR ✗ “Asterks” (parse: KICK ASTERKS) MinerU ✗ “does not provide any evidence” ▼ re-convert, or add the image Unlimited ✓ “AVALAUNCH MEDIA” with image ✓ “Avalaunch Media ... in the banner” DESIGN Treat the parse as fallible: re-convert or attach the image when it is empty or implausible.	Distractor exposure 10 gold pages of a 20-page deck FINDING <i>A few added pages inflate the count; the gold set, or more context, counts right.</i> Q How many lateral bar graphs are present in the document? Gold 10 +3 pages ✗ “there are 12 lateral bar graphs” +5 pages ✗ “there are 13 lateral bar graphs” ▼ remove competing pages, or read on gold pages ✓ “there are 10 lateral bar graphs” +7 to +18 ✓ “there are 10 lateral bar graphs” DESIGN Retrieve broadly for recall, then filter or rerank what reaches the reasoner.	Response calibration two tables, two pages FINDING <i>The question is answerable; the abstain instruction turns multi-step difficulty into refusal.</i> Q How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry? Gold 16.7% none ✓ “... 16.7 percentage points lower” cot ✓ “Step 1: ... Answer: 16.7%” abstain ○ “Not answerable” (false refusal) ▼ reason before deciding abstain_cot ✓ “79.9% − 63.2% = 16.7%” DESIGN Tune refusal jointly with reasoning on the deployed backbone, and report false refusals.